TEMASEK JUNIOR COLLEGE
Preliminary Examinations 2009

BIOLOGY **9747/03**
Higher 2 **Tuesday, 29 September**
Paper 3 **1 hour 30 minutes**

READ THESE INSTRUCTIONS FIRST

Write your name and civics group on all answer papers used.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions

At the end of the examination, fasten the answer papers securely and hand up separately.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
TOTAL	

This document consists of **13** printed pages.

1 Human growth hormone (hGH) is a peptide hormone containing 191 amino acids, synthesized and released by the cells of the anterior pituitary gland. The hormone stimulates growth and cell reproduction in humans. A deficiency of this hormone in children results in growth failure and short stature. In adults, growth hormone deficiency causes decreased muscle mass and poor bone density.

Fig. 1.1 shows how the gene for human growth hormone (hGH) can be transferred into a bacterium.

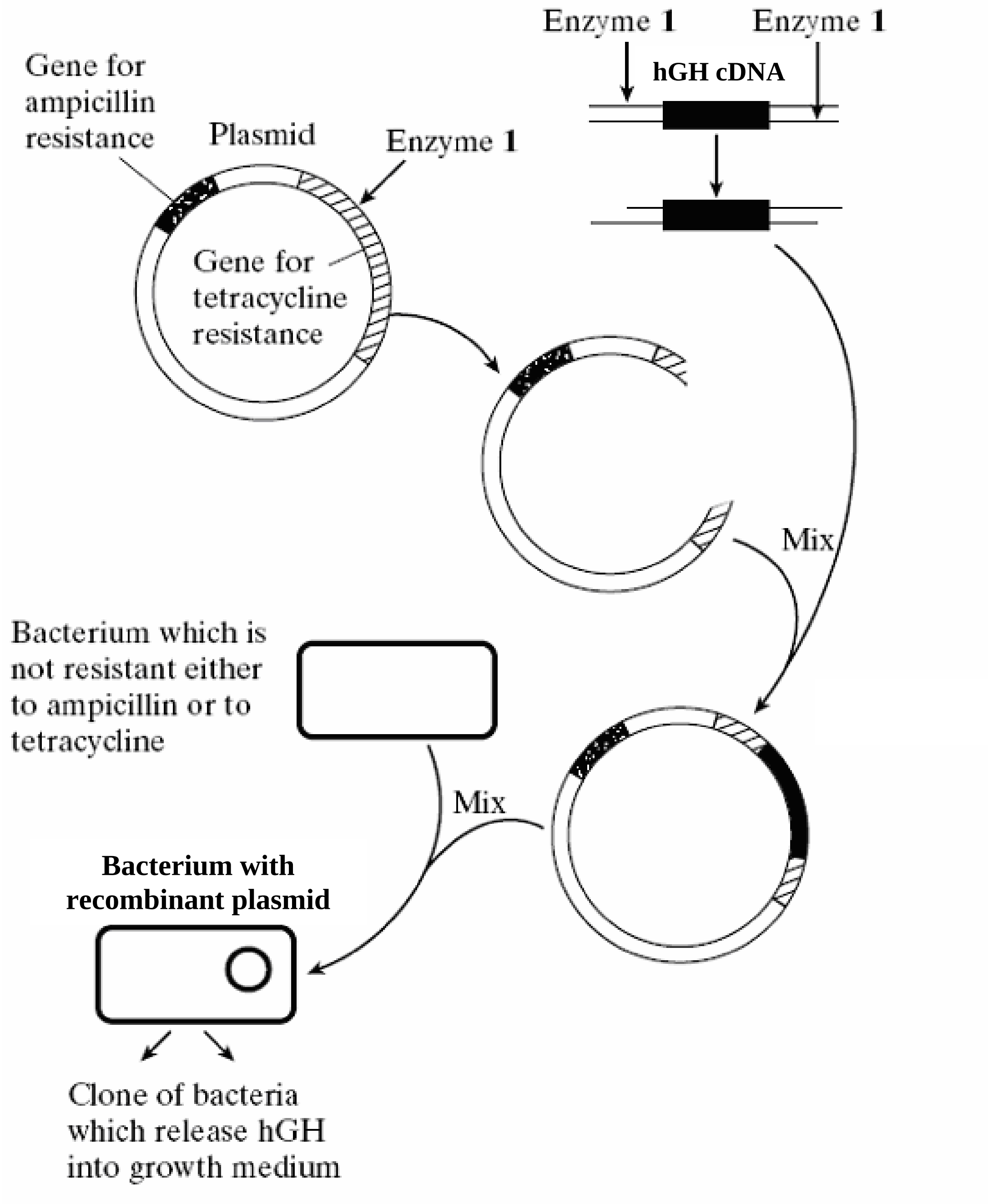


Fig. 1.1

(a) (i) Describe the process by which hGH cDNA is obtained. [3]

- **hGH mRNA is isolated from cells of the anterior pituitary gland. [1]**
- **Reverse transcriptase is added to synthesize a copy of the hGH cDNA. [1]**
- **The mixture is then treated with alkali to digest the mRNA. DNA polymerase is added to make the cDNA double-stranded. [1]**

(ii) Explain why **Enzyme 1** is able to cut both the plasmid DNA and cDNA molecules. [2]

- **Restriction sites for Enzyme 1 are present on both plasmid DNA and cDNA. [1]**
- **The active site of Enzyme 1 has a specific configuration which corresponds to / [is complementary to] the shape of the DNA nucleotide sequence present within the restriction site. [1]**

(iii) Describe the steps involved in introducing the recombinant plasmid into the bacterium. [3]

- **Bacteria cells are made competent by addition of Ca^{2+} in the form of $CaCl_2$ solution. [1]**
- **Plasmid DNA solution is then added to the bacterial cells. [1]**
- **The bacteria cells are then subjected to a brief heat shock at 42°C. This causes holes to appear transiently in the bacterial cell membrane, allowing DNA molecules to enter. [1]**

(iv) Explain **one** possible danger of using plasmids which contain antibiotic resistance genes in genetic engineering. [1]

- **Antibiotic resistance genes may then be passed to other potentially harmful / pathogenic bacteria in the environment. [1]**

Phage viruses are commonly used as a means to transfer foreign DNA into bacteria cells. When a phage infects a layer or lawn of bacterial cells that are growing on a flat surface, a zone of lysis or growth inhibition can occur. This produces a clear zone in the bacterial lawn, known as a plaque. Each plaque originates from a single phage particle.

Fig. 1.2 illustrates the process involved in bacterial transformation and selection using phage vectors.

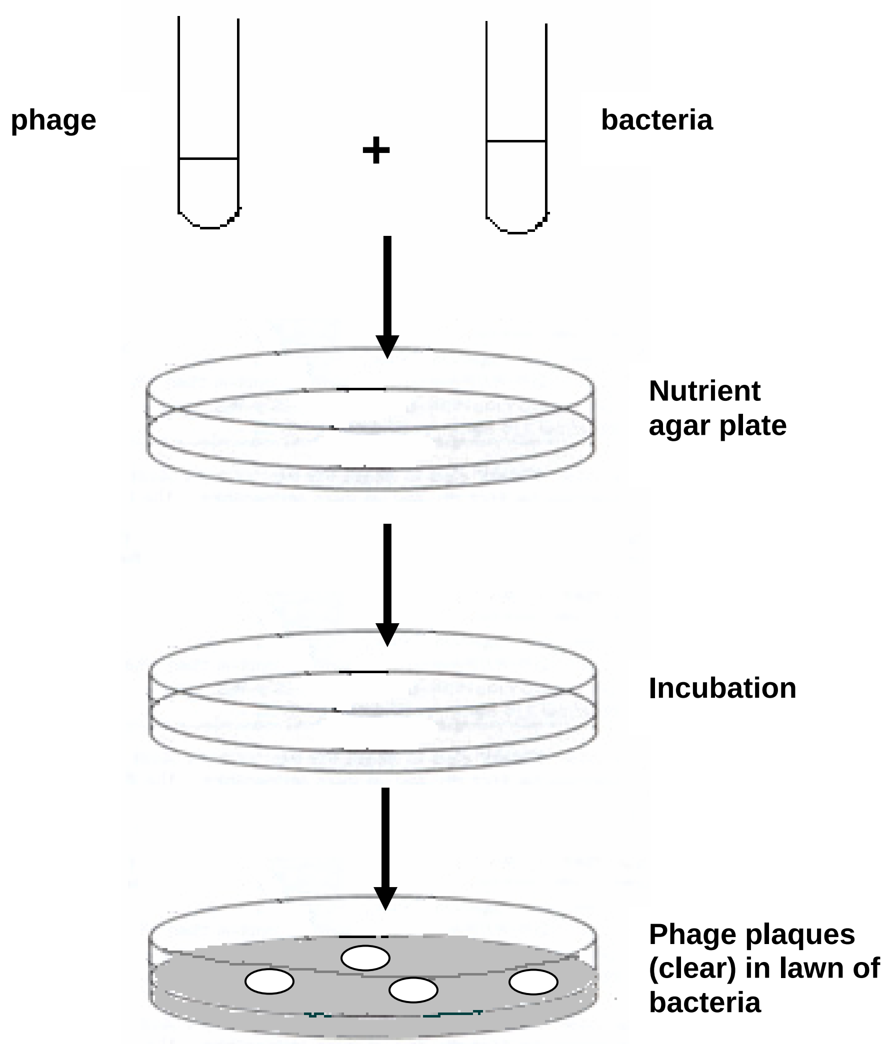


Fig. 1.2

(b) **(i)** With reference to a specific example, describe how a virulent phage can be used as a vector for gene cloning in bacteria. [4]

T4 bacteriophage

- **Genome of the T4 bacteriophage is modified / a part of the viral genome critical for viral replication is deleted. [1]**
- **Both the viral DNA and the gene of interest are cut with the same restriction enzyme, and joined covalently by DNA ligase to form a recombinant phage DNA molecule. [1]**
- **The recombinant DNA molecule is then packaged into capsids and introduced into a host bacteria cell. [1]**
- **Once inside the bacteria cell, the recombinant phage DNA is replicated and transcribed, thus producing more copies of the gene of interest and new phage proteins. [1]**

(ii) Suggest **two** advantages of the use of phage vectors over plasmid vectors in transferring DNA into bacteria cells. [2]

(Any two)

- **Phage vectors can accommodate large DNA inserts [1] (of up to 40 kb), whereas plasmid vectors can only take DNA inserts of up to 10 kb.**
- **Higher transformation efficiency (about 1000 times more efficient than the plasmid vector) as phage viruses specifically target bacteria cells / lower copy no. required of both phage vector and gene of interest. [1]**
- **Temperate phages have the ability to integrate their genome, including the gene of interest, into the bacterial chromosome. Thus, the gene of interest is replicated together with the bacterial chromosome during DNA replication. [1] This ensures each daughter cell contains the gene of interest.**

[Total: 15]

2 Plants have been used for centuries as medicines - now genetically modified plants can produce plastic, skin tissue agents and human blood proteins. Biotechnologists have created genetically modified (GM) plants that can grow plastic. Conventional plastics are made from oil and do not degrade easily, but the plant plastic is biodegradable. To create these plants, four genes from plastic-producing bacteria were inserted into varieties of oilseed rape (Eurasian plant) and cress (as shown in Figure 2.1). Meristematic cells (i.e. undifferentiated cells found in zones where cell division and growth can take place) were targeted for the insertion of the desired genes after many failed attempts using various other cell types.

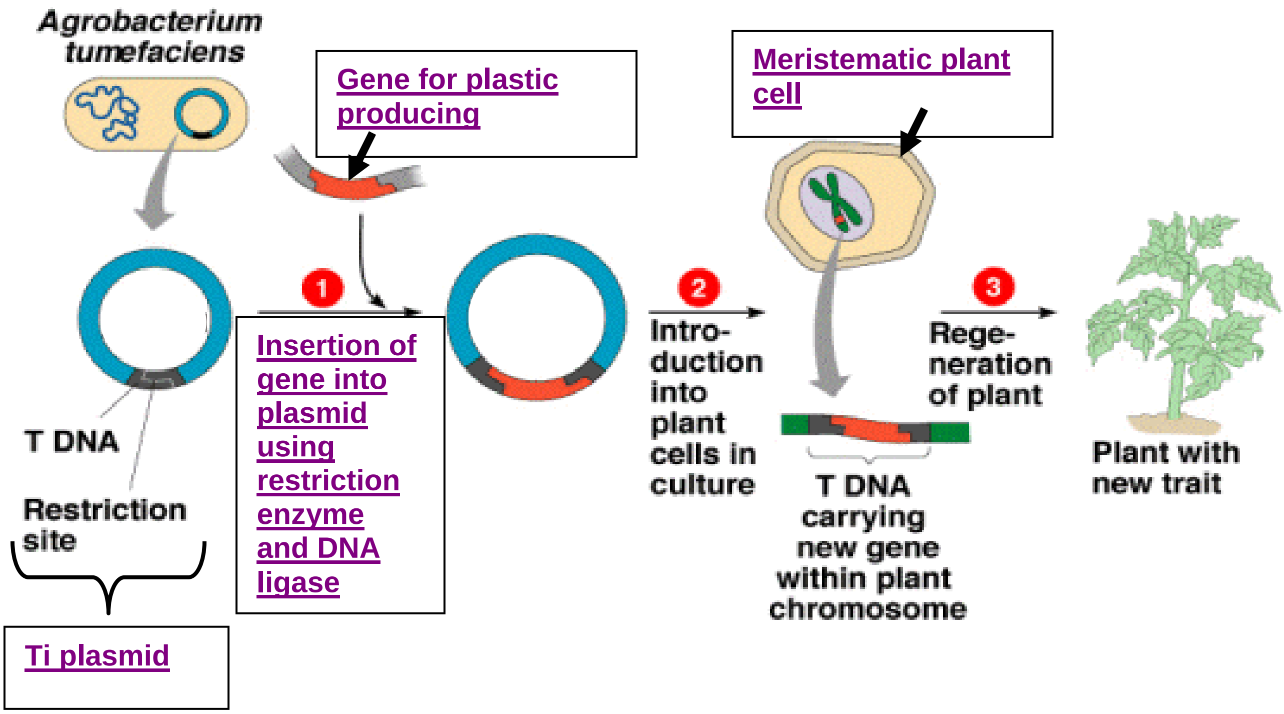


Figure 2.1

(a) Outline the procedure for creating the plastic-producing plants by completing the boxes provided in Figure 2.1. [4]

(b) Selection for transformed plant cells was achieved through observation of a blue coloration in the cell masses. With reference to Figure 2.1, state when and explain how selection was carried out. [2]

- **Between Steps 2 and 3/ immediately after Step 2/ just before Step 3**
- **The Ti plasmid contains a gene for an enzyme called B-glucuronidase (GUS gene).**
- **By growing the plant cell(s) (masses) in the substrate, X-Gluc for B-glucuronidase**
- **A blue precipitate will be formed at the sites of the enzyme reaction indicating the presence of transformed cells.**

(c) Explain why using meristematic cells resulted in a higher success rate. [1]

(Either one:)
- **Thinner cell walls – easier entry of recombinant plasmids**
- **Actively dividing to result in daughter cells with desired genes**

Golden Rice, a variety of *Oryza sativa* rice, produced through genetic engineering to biosynthesize beta-carotene, was developed as a fortified food to be used in areas where there is a shortage of dietary vitamin A. The *psy* (from daffodils) and *crt1* (from a soil bacterium) genes were transformed into the rice nuclear genome and placed under the control of an endosperm specific promoter, so that they are only expressed in the endosperm. Upon successful transformation, the plant cells are cultured as shown in Figure 2.2.

In 2005 a new variety called *Golden Rice 2* was announced which produces up to 23 times more beta-carotene than the original variety of golden rice. Neither variety is currently available for human consumption.

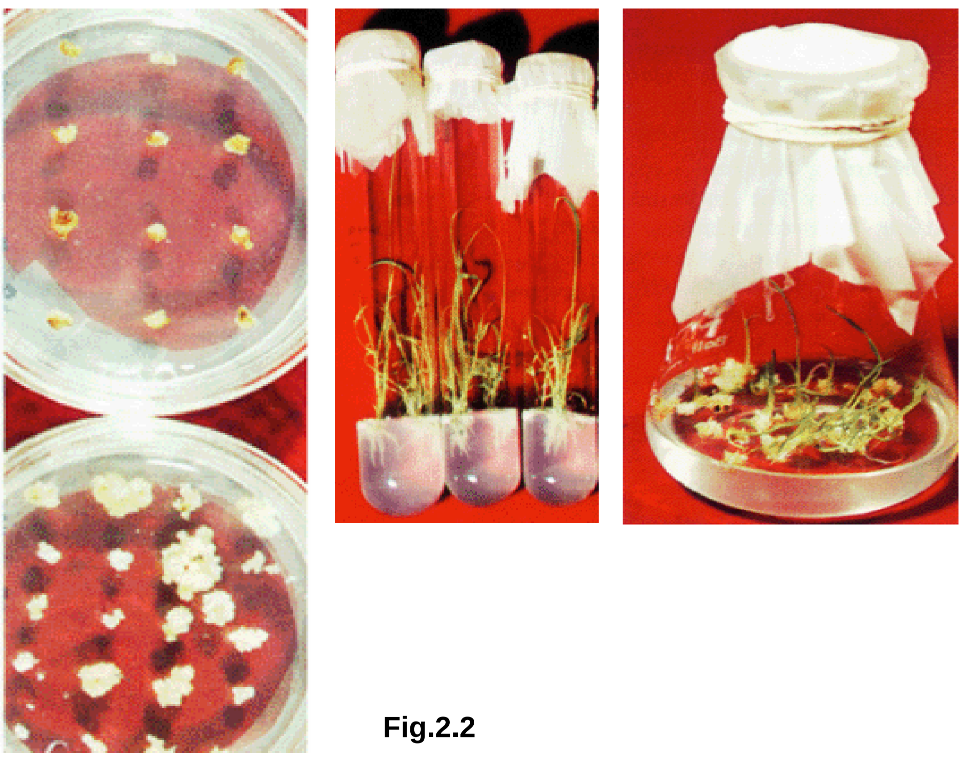


Fig.2.2

(d) State **one** advantage and **one** disadvantage of using the technique shown in Figure 2.2 for the production of the Golden Rice. [2]

(Any one advantage:)

(i) **The production of exact copies of plants that produce beta-carotene.**

(ii) **Possible to create large numbers of clones from a single transformed plant cell.**

(iii) **To quickly produce mature rice plants – reproduction of plants is possible without having to wait for the onset of seed production.**

(iv) The production of multiple plants in the absence of seeds or necessary pollinators to produce seeds.

(v) Overcomes seasonal restrictions for germination.

(vi) The regeneration of whole plants from plant cells that have been genetically modified.

(vii) The production of plants in sterile containers that allows them to be moved with greatly reduced chances of transmitting diseases, pests and pathogens (e.g. to allow for international exchange of sterilized plant materials, eliminating the need for quarantine).

(viii) Selection of desirable traits is possible directly from the culturing setup (*in vitro*), decreasing the amount of space required for field trials.

(Any one disadvantage:)

(i) May not be as convenient as sowing seeds if very large numbers of plants are required, as it is very labour intensive and costly.

(ii) Since all clones are genetically identical, any change in environmental conditions, or the appearance of a new disease, could have devastating consequences if the plants are not resistant or cannot adapt.

(iii) Since the conditions were controlled and optimized, there will be problems in the transition of the plant from medium to soil. Success rate of transplanted individuals is not high.

(iv) It must be carried out in sterile conditions, which requires highly-trained staff and imposes severe constraints on working practice.

(e) Explain why "neither variety is currently available for human consumption". [2]

Concerns:

1. The use of vectors which confer antibiotic-resistance

- **Once these transgenic plants are eaten, the gene for antibiotic resistance may pass from the plant to the *E.coli* bacteria found in the gut, making them resistant to the antibiotic.**

OR

- **The gene may then be passed to other potentially harmful bacteria in the environment which would be antibiotic-resistant.**

2. Gene taken from the soil bacterium – may have undiscovered side effects upon consumption of the rice.

Initial research during the production for Golden Rice involved using *psy* and *crt1* genes obtained through polymerase chain reaction. However, these fragments had blunt ends and thus the concentration of DNA ligase had to be increased to achieve more recombinant plasmids.

(f) Suggest **one** possible advantage of using blunt end fragments for cloning. [2]

(any one:)

- **Overcomes the problem of fragments being inserted in the reverse direction when sticky end fragments are used, since most sticky ends generated have [palindromic symmetry/sequence/OWTTE]**

- **Prevents reannealing of fragments, which occurs spontaneously for sticky end fragments**

- **Plasmid and gene can be cut with different restriction enzymes that produce blunt ends – no need to search for similar restriction sites**

(g) Besides increasing the concentration of DNA ligase, suggest **one** other way to overcome the problem of using blunt end fragments in cloning. [2]

- **Add a short sequence (linker) of either side of the blunt end fragment using artificial synthesis and DNA ligase**
- **Cut both ends with a restriction enzyme that generates sticky ends.**

OR
Use of terminal transferase to add poly-G tail to 3' terminus of gene of interest and to add poly-C tail to 3' terminus of cut plasmid/ vector

[Total: 15]

3 Restriction fragment length polymorphisms (RFLP) have been used in genomic mapping since the 1980s.

(a) Define restriction fragment length polymorphism. [2]

- **Differences in nucleotide sequence between alleles at a locus that result in the formation of restriction fragments of different length [1]**
- **upon digestion by the same restriction enzyme(s) and detection by Southern blot analysis. [1]**

(b) Suggest **two** ways in which RFLPs might be useful in genomic mapping. [2]

Any 2

- **Act as genetic markers/ provide increased number of markers besides gene markers [1]**
- **RFLPs are usually found in non-coding regions.**
  - **Using RFLPs as markers allows mapping of non-coding regions [1]**
  - **Also allows mapping of regulatory sequences [1]**
- **Length of restriction fragments formed allows estimation of number of nucleotides between markers/ genes. [1]**
- **Allows linkage mapping to be carried out, hence determining relative order of genes/ genetic markers on a chromosome. [1]**
- **Any other valid point**

(c) In a study of gene **X** in mice, some scientists decided to use two well-studied RFLP loci, RFLP locus **A** and RFLP locus **B**, to determine the relative position of gene **X** on chromosome 1. Figure **3.1a** shows the relative positions of the two RFLP loci on the chromosome 1, and Figure **3.1b** shows the alleles present at each RFLP locus.

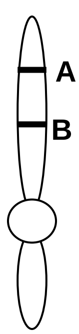


Fig 3.1a

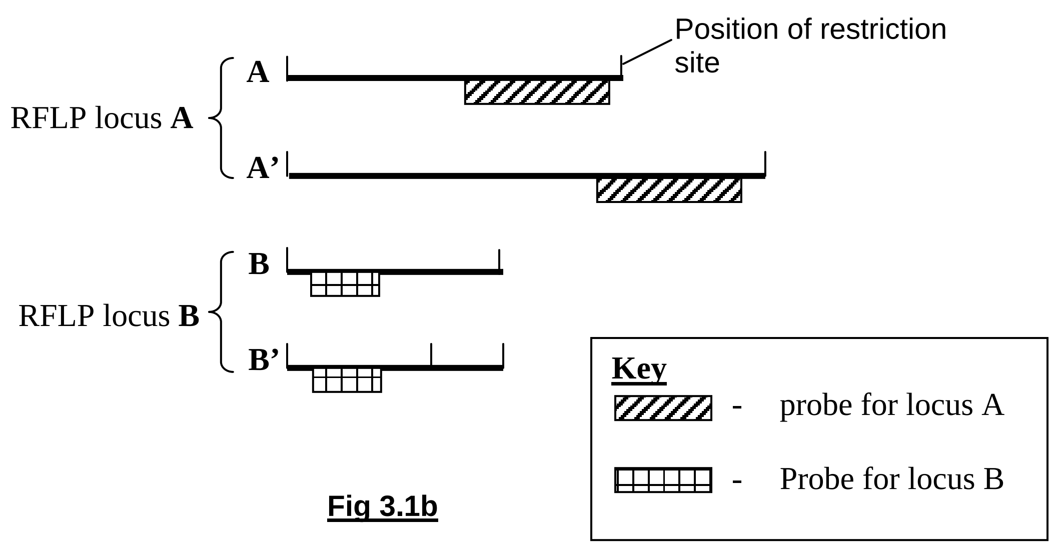


Fig 3.1b

(i) With reference to fig 3.1b, suggest how RFLPs might arise. [2]

- **RFLP locus A: Chromosomal [duplication/ deletion] leading to different numbers of tandem repeats. [1]**
- **RFLP locus B: [Substitution/ deletion/ insertion] resulting in [gain/ loss] of a restriction site. [1]**

To determine the recombination frequency between RFLP loci **A** and **B**, the researchers carried out a few crosses. In cross **1**, mice homozygous for alleles **A** and **B** were crossed with mice homozygous for alleles **A'** and **B'**. Figure **3.2** shows the autoradiograph from mice homozygous for alleles **A** and **B**.

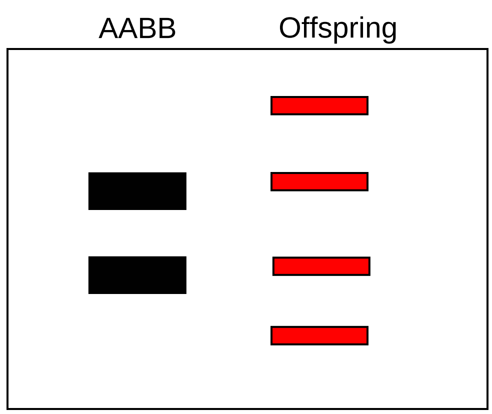


Figure 3.2

(ii) Explain the term recombination frequency. [2]

- **Frequency at which crossing over occurs between two chromosomal loci. [1]**
- **A measure of genetic distance between two loci on a chromosome/ OWTTE. [1]**

(iii) On Fig 3.2, draw the positions of the bands you would expect to see upon analysing the DNA of the offspring from cross **1** above are analysed. [1]

- **Correct position and size [1]**

(iv) Figure **3.3** shows one possible type of gamete that the offspring from cross **1** can form. Complete the diagram by showing all other possible type of gametes that can be formed by the offspring from cross **1**. [1]

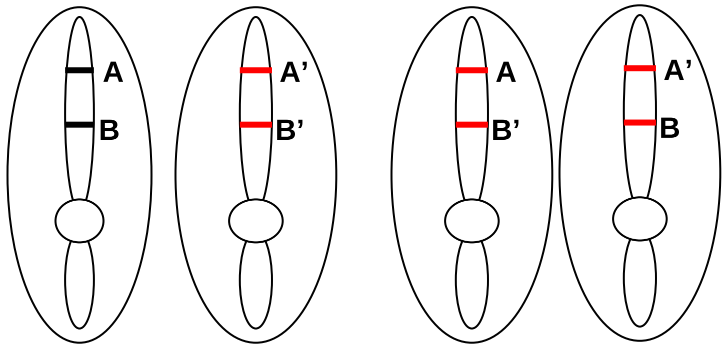


Figure 3.3

In cross **2**, the offspring from cross **1** were crossed with mice homozygous for alleles **A** and **B**. RFLP analysis of the offspring from cross **2** was carried out, and four distinct RFLP profiles were obtained. (shown in figure **3.4**)

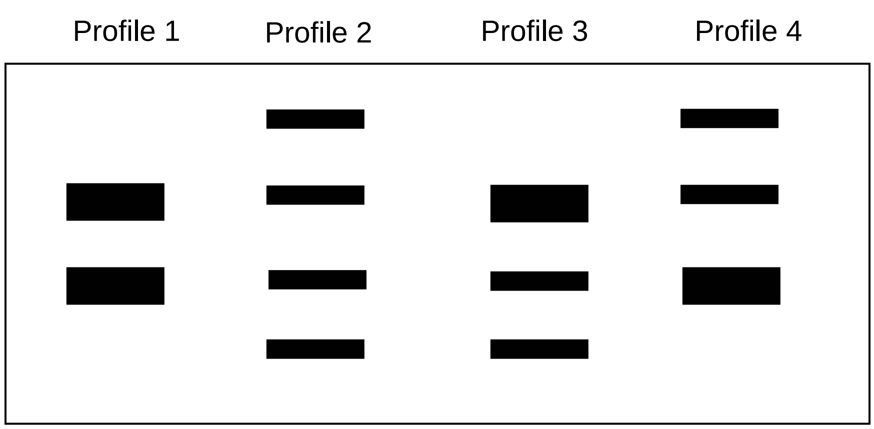


Figure 3.4

The number of mice with each RFLP profile is as follows:

Profile	No. of mice
1	464
2	471
3	103
4	97

(v) Calculate the recombination frequency of RFLP loci **A** and **B**, using the given formula, and show your working in the space provided. [2]

Recombination frequency = total no. of recombinant offspring/ total no. of offspring.

=total no. of mice with profiles 3 and 4/ total no.
= (103+97)/ 1135 [1]
= 0.176 OR 17.6% [1]

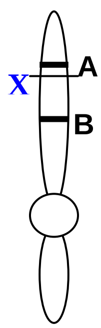
Note: 1m for realising which offspring are recombinant (working), 1m for ans.

In two separate sets of crosses, researchers worked out the recombination frequencies between gene **X** and RFLP locus **A**, as well as that of gene **X** and RFLP locus **B**. These recombination frequencies are given below:

Recombination frequency between **A** and **X**: 3.4%
Recombination frequency between **B** and **X**: 14.2%

(vi) In the diagram given below, indicate the relative position of gene **X**. [1]

- **X located between A and B, X is nearer to A than it is to B [1]**



(d) State **two** other applications of RFLP analysis. [2]
- **DNA fingerprinting/ associated applications of DNA fingerprinting [1]**
- **Disease detection [1]**

[Total: 15]

4 (a) Relate the features of adult stem cells to their use in the treatment of leukemia. [8]

- Adult stem cells (ASC) are undifferentiated cells present in differentiated tissues or organ of both adult and children. [1]

- ASC can renew and divide for long periods of time and can differentiate to form major specialised types of tissue or organ. [1]

Either example 1

Example 1: Bone Marrow Transplant using donor hematopoietic stem cell

- Hematopoietic stem cells are multipotent ASC found in the bone marrow, responsible for constant renewal of blood, giving rise to all blood cells types. e.g. red blood cells and white blood cells. [1]

- In bone marrow transplant, hematopoietic stem cells from a compatible donor are used [1] as treatment for leukemia.

- Chemotherapy is used to kill cancerous white blood cells in patients [1] with leukemia.

- Donor hematopoietic stem cells from bone marrow are isolated [1], and made to differentiate into desirable cell type(s).

- The differentiated cells are then injected into the tissue to replace damaged or non-functional cells [1]

- Transplanted bone marrow contains healthy stems cells which will reproduce to replenish normal blood cells. [1]

Alternative questions for post mortem discussion:

Example 2: Skin grafting

- Skin stem cells are found in the basal layer of the epidermis and at the base of hair follicles. [1]
- epidermal stem cells give reise to keratinocytes [1], which migrate to the surface of the skin and form a protective layers.
- Follicular stem cells give rise to both the hair follicle and the epidermis.[1]
- Skin cells can be grown in large numbers, providing life-saving skin grafting treatment for burn victims. [1]

Example 3. Diabetes treatment

- Diabetes might be treated by generating insulin producing cells from stem cells obtained from a normal pancreas and transplanting them to the pancreas of a diabetes patient. [1]

(b) Using specific example(s), compare viral and non-viral delivery systems in the treatment of cystic fibrosis. [6]

Introductory statement:

- Viral delivery systems for treatment of cystic fibrosis involves the use of adenoviruses and adeno-associated viruses; Non-viral delivery system for treatment of cystic fibrosis involves use of liposomes OR gene gun.

Similiarities:

- Both adeno-associated viruses and liposomes do not produce immune response in the body. [1]
- Both adenoviruses and non-viral do not result in long term expression of the gene and these methods does not result in integration of normal gene into host cell genome. [1]

Differences:

Basis of comparison	Viral methods	non-viral methods	
effectiveness of gene transfer	• adenoviruses naturally infect lungs, and hence are effective for delivering the gene	• both liposome and gene gun are less effective	[1]

long term expression of gene	• use of adeno-associated virus can result in integration of normal gene into host cell genome, and may result in long term expression of gene.	• low	[1]
immune response	• use of adenoviruses can cause body to produce immune response	• minimal host immune response is generated	[1]
side effects	• not known	• minimal	[1]

Note: specific references to the particular viral and non-viral methods must be made during comparison. different viral or non-viral methods cannot be grouped together in the same comparison if the point discussed is not common for all viral or non-viral methods discussed.

(c) Describe the process of Restriction Fragment Length Polymorphism (RFLP) detection.[6]

- Definition of RFLP [1]
- [PCR DNA sample to obtain enough copies of DNA] – not a marking
- <u>DNA</u> sample is first <u>digested with restriction enzymes and separated by agarose gel electrophoresis</u>.[1]
- DNA in the gel <u>are denatured into single-strands by addition of NaOH</u> and <u>transferred to nitrocellulose membrane by capillary action.</u> [1]
- <u>Nitrocellulose membrane</u> is <u>baked</u> to <u>permanently cross-link the single-stranded DNA to the membrane.</u> [1]
- The <u>membrane is then incubated with radioactively labelled DNA or RNA probe</u> which is complementary to the gene of interest / RFLP loci, the <u>hybridised probe is detected by autoradiography</u>. [1]
- The nitrocellulose membrane will shows a band of radioactivity where the probe hybridised with complementary gene of interest/RFLP loci.[1]

Note: 1 additional mark bonus.

End of Paper